

A Comprehensive Survey of Topological-Space Set Theoretic Estimation

with an eye to mechanization

Project `thejeb`

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Abstract

Following the two seed papers [4, 5], this survey maps the wider body of work in which set theoretic estimation (STE) lives — the convex feasibility problem and its projection algorithms, the topological and nonconvex generalizations, and the set-membership tradition in control — and, crucially, the current state of *machine-checked* convex analysis in Lean 4 / Mathlib. The purpose is to prioritize what `thejeb` mechanizes next. Statements here are attributed to their sources; nothing is claimed proven in our library unless it appears in the CI-checked `Ste` development.

1 The core problem STE instantiates

STE’s mathematical heart is the *convex feasibility problem* (CFP): given closed convex sets $C_1, \dots, C_m$ (or countably many) in a Hilbert space H , find a point of $\bigcap_i C_i$ [1, 5]. Combettes’ feasibility set is exactly this intersection; our `Ste.Convex` proves the intersection convex, the geometric precondition for everything below. The CFP is a modeling paradigm across image recovery, tomography, sensor networks, optimal/robust control, and machine learning [1, 6].

2 Projection algorithms (the convex engine)

POCS and cyclic projections. The method of successive projections onto convex sets [11], descended from Bregman’s relaxation and cyclic projection method [3], converges (weakly, in general) to a feasible point when the intersection is nonempty. The unifying convergence theory — averaged/extrapolated projections, control strategies for set selection, and (bounded) linear regularity giving linear rates — is surveyed definitively by Bauschke and Borwein [1] and set within monotone operator theory by Bauschke and Combettes [2].

Splitting and Douglas–Rachford. Beyond plain projections, proximal splitting [7] and the Douglas–Rachford (DR) reflection method extend feasibility solving to sums of operators. DR is empirically the strongest projection method and enjoys a rich convergence theory; the sixty-year survey of Lindstrom and Sims [9] is the entry point.

3 Leaving convexity: the topological turn

Carlson’s move to a *topological* solution space [4] parallels a broader thread: once the constraint sets are not convex, Hilbert-space nonexpansiveness arguments fail and convergence must be re-

covered from *regularity/transversality* conditions (superregularity, linear regularity) rather than convexity [9]. Global DR convergence is known for structured nonconvex cases (a sphere with a line/hyperplane; finite unions of convex sets, locally). Carlson’s motivation is complementary: he adopts a topological space precisely so that property sets are *static* a priori information that never expand, sidestepping the Optimal Bounding Ellipsoid pathology of metric-space STE, in which the bounding volume can spuriously enlarge the solution set. `Ste.Topology` captures the well-posedness half of this turn (closedness; compactness + finite intersection property $\Rightarrow$ nonempty feasibility), which specializes to Carlson’s finite key spaces.

4 The set-membership tradition (control)

Running in parallel to signal-processing STE is *set-membership / unknown-but-bounded* estimation in control [8, 10]: the feasible *state* set consistent with bounded-error measurements, propagated recursively and outer-bounded by ellipsoids or zonotopes. This is exactly STE with a dynamical model, and it is where the OBE algorithms Carlson critiques originate. It remains active for safety-critical and robust/adaptive control (fault detection as feasibility, robust MPC over uncertainty sets). For `thejeb` it is a source of concrete, finitely-describable property sets (halfspaces, boxes, ellipsoids) that would make good mechanized examples.

5 State of machine-checked convex analysis (the tooling frontier)

This is the decisive input for our roadmap. Mathlib already carries Hilbert-space inner-product geometry and projection onto closed convex sets. Beyond the library, recent Lean 4 formalizations have reached exactly the algorithms STE needs:

- convergence of the Krasnosel’skiĭ–Mann and Halpern fixed-point iterations, and other fixed-point algorithms in Hilbert spaces (arXiv:2602.17064, “Formalization of Two Fixed-Point Algorithms in Hilbert Spaces”);
- convergence *rates* of first-order convex optimization algorithms, with gradient/subgradient theory for convex functions on a Hilbert space (*J. Automated Reasoning*, 2025, “Formalization of Convergence Rates of Four First-order Algorithms for Convex Optimization”; see also arXiv:2403.11437).

Because POCS and DR are fixed-point iterations of (firmly) nonexpansive projection/reflection operators, these results are the scaffolding for a machine-checked STE convergence theorem on top of our `Ste.Convex`. Isabelle/HOL-Analysis (Carathéodory, Radon, Helly) and Coq polyhedra formalizations provide comparison points but are off our Lean path.

6 Prioritized mechanization roadmap

1. **POCS convergence** for finitely many closed convex C_i with nonempty intersection, as a fixed-point iteration atop `Ste.Convex`, reusing Mathlib projection theory and the Krasnosel’skiĭ–Mann formalization. *Highest value: turns our static feasibility geometry into a verified solver.*
2. **Carlson cipher counting** (Lemma 4.1, Theorem 4.1) over `Fin` alphabets — self-contained finite combinatorics, no analysis; good parallel track.

3. **Cipher reduction** (Theorem 5.2, Corollary 5.3): every block cipher is a block substitution cipher — algebra of keyed permutations.
4. **Regularity-based results** and Douglas–Rachford [9]: reach for these only after POCS, as they need linear/super-regularity infrastructure not yet in Mathlib.
5. **AEP typical sets** (Carlson Theorem 2.3): deepest; needs a probability/information-theory layer.

Method note

Consistent with the project rules, this survey looks to seminal work (Bregman, Schweppe, Youla–Webb, Combettes, Bauschke) and considers the field broadly, but treats every mathematical claim as unverified for our purposes until it is machine-checked in the `Ste` library. Open targets are tracked in `log/`.

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